

Frontier LLM Cluster
(Claude 3.5 Sonnet / Opus,
GPT-4o, Gemini 3.1 Pro)
Generates Candidate y



SuperGravity Guard
AST Anti-Stub Scanner
Rejects 'pass', '...', mocks
Fail-Closed Attestation



Sandbox Execution
Native rustc -O / NumPy
Physical Resource Metering
Duration (ms) & RAM (MB)



Physical Invariant Gate
 $I(s) = 0$ Verification
 $\epsilon_{\text{inv}} \leq \epsilon_{\text{tol}}$
Proof Token Minting

Stub / Failure



Thermodynamic Hardness Penalty
If Exception / Stub / Violation:
 $E(y) = 10^6$ (Maximum Pain)
Immediate Rejection Wall

Verified Token



Direct Preference Optimization (DPO)
 $R(y) = -E(y)$
Pairwise Margin: $\Delta R = R(y_w) - R(y_l) \geq 3.023$
Empirical Loss Reduction: -20.1%